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Missing values in longitudinal data

13.1 Introduction

Missing values arise in the analysis of longitudinal data whenever one or more of the sequences of measurements from units within the study are incomplete, in the sense that *intended* measurements are not taken, are lost, or are otherwise unavailable. The emphasis is important: if we choose in advance to take measurements every hour on one-half of the subjects and every two hours on the other half, the resulting data could also be described as incomplete but there are no missing values in the sense that we use the term; we call such data *unbalanced*. This is not just playing with words. Unbalanced data may raise technical difficulties – we have seen that some methods of analysis can only cope with data for which measurements are made at a common set of times on all units. Missing values raise the same technical difficulties, since of necessity they result in unbalanced data, but also deeper conceptual issues, since we have to ask *why* the values are missing, and more specifically whether their being missing has any bearing on the practical questions posed by the data.

A simple (non-longitudinal) example makes the point explicitly. Suppose that we want to compare the mean concentrations of a hormone in blood samples taken from ten subjects in each of two groups: one a control, the other an experimental treatment intended to suppress production of the hormone. We are presented with ten assayed values of the hormone concentration from the control subjects, eight assayed values from the treated subjects, and are told that the values from the other two treated subjects are 'missing'. If we knew that this was because somebody dropped the test tubes on the way to the assay lab, we would probably be happy to proceed with a two-sample t-test using the 18 non-missing values. If we knew that the missing values were from subjects whose hormone concentrations fell below the sensitivity threshold of the assay, ignoring the missing values would mask the very effect we were looking to detect. Lest this example offend the sophisticated reader, we suggest that more subtle versions of it are not uncommon in real longitudinal studies. For example, it may be that a sequence of atypically low (or high) values on a particular unit foreshadows its removal from the study.

13.2 Classification of missing value mechanisms

Little and Rubin (1987), give a general treatment of statistical analysis with missing values, which includes a useful hierarchy of missing value mechanisms. Let $\boldsymbol{Y}^*$ denote the complete set of measurements which would have been obtained were there no missing values, and partition this set into $\boldsymbol{Y}^* = (\boldsymbol{Y}^{(o)}, \boldsymbol{Y}^{(m)})$ with $\boldsymbol{Y}^{(o)}$ denoting the measurements actually obtained and $\boldsymbol{Y}^{(m)}$ the measurements which would have been available had they not been missing, for whatever cause. Finally, let $\boldsymbol{R}$ denote a set of indicator random variables, denoting which elements of $\boldsymbol{Y}^*$ fall into $\boldsymbol{Y}^{(o)}$ and which into $\boldsymbol{Y}^{(m)}$. Now, a probability model for the missing value mechanism defines the probability distribution of $\boldsymbol{R}$ conditional on $\boldsymbol{Y}^* = (\boldsymbol{Y}^{(o)}, \boldsymbol{Y}^{(m)})$. Little and Rubin classify the missing value mechanism as

- *completely random* if $\boldsymbol{R}$ is independent of both $\boldsymbol{Y}^{(o)}$ and $\boldsymbol{Y}^{(m)}$
- *random* if $\boldsymbol{R}$ is independent of $\boldsymbol{Y}^{(m)}$
- *informative* if $\boldsymbol{R}$ is dependent on $\boldsymbol{Y}^{(m)}$.

It turns out that for likelihood-based inference, the crucial distinction is between random and informative missing values. To see this, let $f(\boldsymbol{y}^{(o)}, \boldsymbol{y}^{(m)}, \boldsymbol{r})$ denote the joint probability density function of $(\boldsymbol{Y}^{(o)}, \boldsymbol{Y}^{(m)}, \boldsymbol{R})$ and use the standard factorization to express this as

$$f(\boldsymbol{y}^{(o)}, \boldsymbol{y}^{(m)}, \boldsymbol{r}) = f(\boldsymbol{y}^{(o)}, \boldsymbol{y}^{(m)}) f(\boldsymbol{r} \mid \boldsymbol{y}^{(o)}, \boldsymbol{y}^{(m)}). \tag{13.2.1}$$

For a likelihood-based analysis, we need the joint pdf of the observable random variables, $(\boldsymbol{Y}^{(o)}, \boldsymbol{R})$, which we obtain by integrating (13.2.1) to give

$$f(\boldsymbol{y}^{(o)}, \boldsymbol{r}) = \int f(\boldsymbol{y}^{(o)}, \boldsymbol{y}^{(m)}) f(\boldsymbol{r} \mid \boldsymbol{y}^{(o)}, \boldsymbol{y}^{(m)}) \mathrm{d}\boldsymbol{y}^{(m)}. \tag{13.2.2}$$

Now, if the missing value mechanism is random, $f(\boldsymbol{r} \mid \boldsymbol{y}^{(o)}, \boldsymbol{y}^{(m)})$ does not depend on $\boldsymbol{y}^{(m)}$ and (13.2.2) becomes

$$\begin{aligned} f(\boldsymbol{y}^{(o)}, \boldsymbol{r}) &= f(\boldsymbol{r} \mid \boldsymbol{y}^{(o)}) \int f(\boldsymbol{y}^{(o)}, \boldsymbol{y}^{(m)}) \mathrm{d}\boldsymbol{y}^{(m)} \\ &= f(\boldsymbol{r} \mid \boldsymbol{y}^{(o)}) f(\boldsymbol{y}^{(o)}). \end{aligned} \tag{13.2.3}$$

Finally, taking logarithms in (13.2.3), the log-likelihood function is

$$L = \log f(\boldsymbol{r} \mid \boldsymbol{y}^{(o)}) + \log f(\boldsymbol{y}^{(o)}), \tag{13.2.4}$$

which is maximized by separate maximization of the two terms on the right-hand side. Since the first term contains no information about the distribution of $\boldsymbol{Y}^{(o)}$, we can ignore it for the purpose of making inferences about $\boldsymbol{Y}^{(o)}$.

Because of the above result, both completely random and random missing value mechanisms are sometimes referred to without distinction as *ignorable*. However, it is important to remember that 'ignorability' in this sense relies on the use of the likelihood function as the basis for inference. For example, the method of generalized estimating equations, as described in Section 8.2.3, is valid only under the stronger assumption that the missing value mechanism is completely random. Also, even within a likelihood-based analysis, the treatment of a random missing value mechanism as ignorable makes several tacit assumptions. The first of these, as emphasized by Little and Rubin, is that $f(\boldsymbol{y}^{(o)})$ and $f(\boldsymbol{r} \mid \boldsymbol{y}^{(o)})$ are separately parameterized, which need not be the case; if there are parameters common to $f(\boldsymbol{y}^{(o)})$ and $f(r \mid \boldsymbol{y}^{(o)})$, ignoring the first term in (13.2.4) leads to a loss of efficiency. Secondly, maximization of the second term on the right-hand side of (13.2.4) implies that the unconditional distribution of $\boldsymbol{Y}^{(o)}$ is the correct inferential focus. Again, this need not be the case; for example, in a clinical trial concerned with a life-threatening condition in which missing values identify patients who have died before the end of the study and $\boldsymbol{Y}^{(o)}$ measures health status, it may be more sensible to make inferences about the distribution of time to survival and the *conditional* distribution of $\boldsymbol{Y}^{(o)}$ given survival rather than about the unconditional distribution of $\boldsymbol{Y}^{(o)}$.

13.3 Intermittent missing values and dropouts

An important distinction is whether missing values occur intermittently or as dropouts. Suppose that we intend to take a sequence of measurements, $Y_1, \ldots, Y_n$, on a particular unit. Missing values occur as *dropouts* if whenever Y_j is missing, so are Y_k for all $k \geq j$; otherwise we say that the missing values are *intermittent*. In general, dealing with intermittent missing values is more difficult than dealing with dropouts because of the wider variety of patterns of missing values which need to be accommodated.

When intermittent missing values arise through a known censoring mechanism, for example, if all values below a known threshold are missing, the EM algorithm (Dempster *et al.*, 1977) provides a possible theoretical framework (Laird, 1988; Hughes, 1999). When intermittent missing values do not arise from censoring, the reason for their being missing is often known, since the subjects in question remain in the study, and in some cases this information will make it reasonable to assume that the missingness is unrelated to the measurement process. In such cases, the resulting data can be analysed by any method which can accommodate unbalanced data. Furthermore, if the method of analysis is likelihood-based, the inferences will be valid under the weaker assumption that the missing value mechanism is random.

In contrast, dropouts are frequently lost to any form of follow-up and we have to admit the possibility that they arise for reasons directly or indirectly connected to the measurement process.

An example of dropout directly related to the measurement process arises in clinical trials, where ethical considerations may require a patient to be withdrawn from a trial on the basis of their observed measurement history. Murray and Findlay (1988) discuss this in the context of long-term trials of drugs to reduce blood pressure where 'if a patient's blood pressure is not adequately controlled then there are ethical problems associated with continuing the patient on their study medication'. Note that a trial protocol which specifies a set of circumstances under which a patient *must* be withdrawn from the trial on the basis of their observed measurement history defines a *random*, and therefore ignorable, dropout mechanism in Little and Rubin's sense.

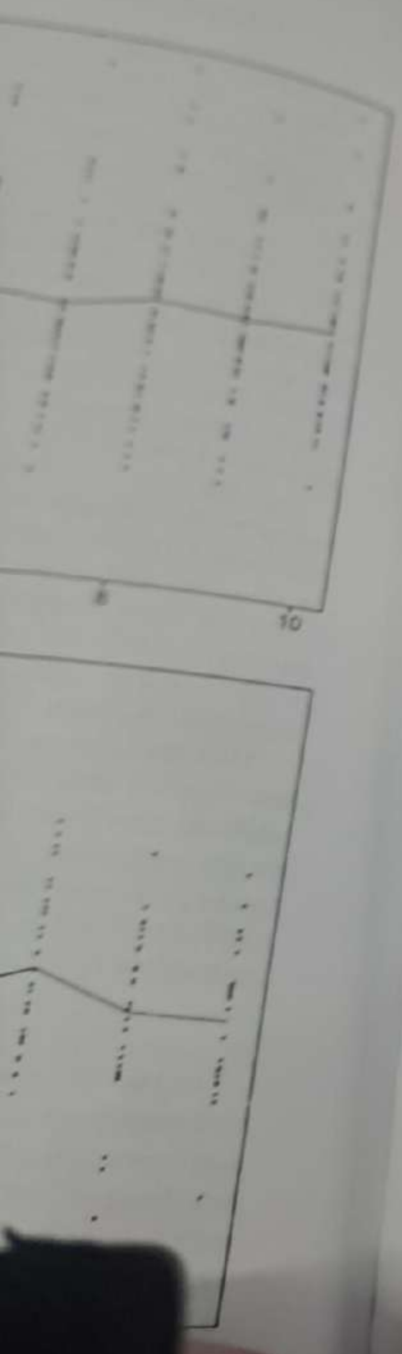
In contrast, the 'dropouts' in the data on protein content of milk samples arose because the cows in question calved after the beginning of the experiment (Cullis, 1994), and there may well be an indirect link between calving date and milk quality.

When there is any kind of relationship between the measurement process and the dropout process, the interpretation of apparently simple trends in the mean response over time can be problematic, as in the following simulated example.

We simulated two sets of data from a model in which the mean response was constant over time, and the random variation within units followed the uniform correlation model described in Section 4.2.1. In each data-set, up to ten responses, at unit time-intervals, were obtained from each of 100 subjects. Dropouts occurred at random, according to the following model: conditional on the observed responses up to and including time $t-1$, the probability of dropout at time t is given by a logistic regression model,

$$\mathrm{logit}(p_t) = -1 - 2y_{t-1}.$$

In the first simulation, shown in Fig. 13.1, the correlation between any two responses on the same unit was $\rho = 0.9$, and the empirical mean responses, calculated at each time-point from those subjects who had not yet dropped out, show a steadily rising trend. A likelihood-based analysis which ignores the dropout process leads to the conclusion that the mean response is essentially constant over time whereas the empirical means suggest a clearly increasing time-trend. There is no contradiction in this apparent discrepancy: the likelihood-based analysis estimates the mean response which would have been observed had there been no dropouts, whereas the empirical means estimate the conditional mean response in the sub-population who have not dropped out by time t. Another way to expla[illegible]iscrepancy between the likelihood-based and empirical estimate[illegible]an response is that the former recognizes the correlation in



...e data and, in effect, imputes the missing values for a particular subject ...ing into account the same subject's observed values. To confirm this, Fig. 13.1(b) shows the result of the second simulation, which uses the same model for measurements and dropouts except that the within-unit correlation is $\rho = 0$. Both the empirical means and a likelihood-based inference now tell the same story – that there is no time-trend in the mean response.

These examples may convince the reader, as they convince us, that it would be useful to have methods for analyzing data with a view to distinguishing amongst completely random, random and informative dropouts. Fortunately, the additional structure of data with dropouts makes this a more tractable problem than in the case of intermittent missing values.

In the rest of this chapter, we concentrate on the analysis of data with dropouts. In Section 13.4 we briefly mention two simple, but in our view inadequate, solutions to the problem. In Section 13.5 we describe methods for testing whether dropouts are completely random. In Section 13.6 we describe an extension to the method of generalized estimating equations which gives consistent inferences under the assumption of random dropouts. In Section 13.7 we review various model-based approaches which can accommodate completely random, random or informative droputs, henceforth abbreviated to CRD, RD and ID respectively.

13.4 Simple solutions and their limitations

13.4.1 *Last observation carried forward*

As the name suggests, this method of dealing with dropouts consists of extrapolating the last observed measurement for the subject in question to the remainder of their intended time-sequence. A refinement of the method would be to estimate a time-trend, either for an individual subject or for a group of subjects allocated to a particular treatment, and to extrapolate not at a constant level, but relative to this estimated trend. Thus, if y_{ij} is the last observed measurement on the ith subject, $\hat{\mu}_i(t)$ is their estimated time-trend and $r_{ij} = y_{ij} - \hat{\mu}_i(t_j)$, the method would impute the missing values as $y_{ik} = \hat{\mu}_i(t_k) + r_{ij}$ for all $k > j$.

Last observation carried forward is routinely used in the pharmaceutical industry, and elsewhere, in the analysis of randomized parallel group trials for which a primary objective is to test the null hypothesis of no difference between treatment groups. In that context, it can be argued that dropout, and the subsequent extrapolation of last observed values, is an inherent feature of the random outcome of the trial for a given subject. Validity of the test for no difference between treatment groups is then imparted by the randomization, without requiring explicit modelling assumptions. ...further argument to justify last observation carried forward is that if, ...ample, subjects are expected to show improvement over the duration ...itudinal trial (i.e. treatment is beneficial), carrying forward a last completely random within each group. We therefore recommend that for preliminary screening, the data should first be divided into homogeneous sub-groups.

Let p_{ij} denote the probability that the ith unit drops out at time t_j. Under the assumption of completely random dropouts, the p_{ij} may depend on time, treatment, or other explanatory variables but cannot depend on the observed measurements, $\boldsymbol{y}_i$. The method developed in Diggle (1989) to test this assumption consists of applying separate tests at each time within each treatment group and analysing the resulting sample of p-values for departure from the uniform distribution on $(0, 1)$. Combination of the separate p-values is necessary if the procedure is to have any practical value, because the individual tests are typically based on very few cases and therefore have low power.

The individual tests are constructed as follows. For each of $k = 1, \ldots, (n-1)$, define a function, $h_k(y_1, \ldots, y_k)$. We will discuss the choice of $h_k(\cdot)$ shortly. Now, within each group and for each time-point t_k, $k = 1, \ldots, n-1$, identify the R_k units which have $n_i \geq k$ and compute the set of scores $h_{ik} = h_k(y_{i1}, \ldots, y_{ik})$, for $i = 1, \ldots, R_k$. Within this set of R_k scores, identify those r_k scores which correspond to units with $m_j = k$, that is, units which are about to drop out. If $1 \leq r_k < R_k$, test the hypothesis that the r_k scores so identified are a random sample from the 'population' of R_k scores previously identified. Finally, investigate whether the complete set of p-values observed by applying this procedure to each time within each treatment group behaves like a random sample from the uniform distribution on $(0, 1)$. The implicit assumption that the separate p-values are mutually independent is valid precisely because once a unit drops out it never returns.

Our first decision in implementing this procedure is how to choose the functions $h_k(\cdot)$. Our aim is to choose these so that extreme values of the scores, h_{ik}, constitute evidence against completely random dropouts. A sensible choice is a linear combination,

$$h_k(y_1, \ldots, y_k) = \sum_{j=1}^{k} w_j y_j. \tag{13.5.1}$$

As with any *ad hoc* procedure, the success of this will be influenced by the investigator's ability to pick a set of coefficients which reflect the actual dependence of the dropout probabilities on the observed measurement history. If dropout is suspected to be an immediate consequence of an abnormally low measurement we should choose $w_k = 1$ and all other $w_j = 0$. If it is suspected to be the culmination of a sustained sequence of low measurements we would do better using equal weights, $w_j = 1$ for all j.

The next decision is how to convert the scores, h_{ik}, into a test statistic and p-value. A natural test statistic is $\bar{h}_k$, the mean of the r_k scores